Course Project

STAT 184 — Summer 2026

Purpose

The purpose of this project is to provide you with an opportunity to put into practice everything that you have learned over the course of the semester and to push yourselves. To this end, you'll work in teams to conduct your own Exploratory Data Analysis, posing and answering your own research question(s).

Past examples

These are from past sections. Their requirements were slightly different and their section headings vary, so **follow the Required report structure below rather than copying their layout**, but they give you a good sense of the expected scope, depth, and quality:

- IPL Batting Performance Analysis (Rohan & Pratham)
- Customer Shopping Trends (Ava, Sydney & Kacie)
- IMDB Movie Data Analysis (Sanjana, Nishita & Jess)

Deadlines

Milestone	Due	Weight (of project grade)
Team formation	Thu Jun 4, 11:59 p.m.	— (required)
Checkpoint 1: topic, question, data sources	Sunday Jun 14, 11:59 p.m.	15%
Oral check-in #1 (mid-project)	Thu Jun 18 in class	10%
Checkpoint 2: partial draft with visualizations	Wed Jun 24, 11:59 p.m.	15%

Milestone	Due	Weight (of project grade)
Oral check-in #2 (final walkthrough)	Thu Jun 25 in class	10%
In-class presentation	Fri Jun 26 (in-class)	20%
Final report	Fri Jun 26, 11:59 p.m.	30%

Teams

- Teams of **2–3 students**.
- Form teams by the deadline above. If you don't have a team by then, I will assign one.
- All team members are individually responsible for understanding the entire project.

Project report guidelines

Topic and questions

- [ ] Pick a topic your team is genuinely interested in. Sports, music, video games, food, news, science, civics — anything is fair game as long as the data exists.
- [ ] Come up with 1–3 clear, answerable research questions.

Data

- [ ] **Your primary data source may *not* be a dataset we used in class.**
- [ ] **Your primary data source may *not* come from an R package.** Supplementary sources are fine from anywhere.
- [ ] Read in your data and perform necessary tidying, wrangling, and cleaning.
- [ ] Describe data provenance: where did it come from, who collected it, why, what are the cases.

Good sources of real data:

- https://data.gov — US government open data
- https://github.com/fivethirtyeight/data — FiveThirtyEight datasets
- https://github.com/nytimes — NYT data
- https://github.com/rfordatascience/tidytuesday — Tidy Tuesday (you may use these CSVs as primary data, just not the R package itself)
- https://scorenetwork.org/ — Sports data
- https://www.kaggle.com/datasets — Kaggle

Analysis and report

- ☐ Conduct an Exploratory Data Analysis.
- ☐ Produce a reproducible report in Quarto, output as **PDF**.
- ☐ Code in a Code Appendix at the end. Use this snippet:

```
{{r codeAppend, ref.label=knitr::all_labels(), echo=TRUE, eval=FALSE}}
```

- ☐ Body of the report should be **narrative + figures + tables**, not code.
- ☐ The body must be **at least 10 pages**. Title page, references, and Code Appendix do **not** count toward the minimum.
- ☐ Include **at least three figures** (plots) and **at least one non-trivial table** (a summary or aggregated table, not a dump of raw data).
- ☐ At least one figure must display **3+ variables** effectively (use color, facets, size, shape, etc.).
- ☐ Every figure and table must have a **caption** and be **cross-referenced by number** from the narrative ("Figure 1 shows…", "as summarized in Table 1…"), with a few sentences of discussion near each.
- ☐ Cite your data sources and any code you adapted from outside (APA, MLA, or footnotes).

Required report structure

> **!** Important
>
> Your report **must** include the following sections, in this order, with these headings (or clearly equivalent ones). Graders look for each section by name. The point values for each appear in the final report rubric.

1. **Title block**: report title, all team members' names, and the date.
2. **Introduction & Motivation**: what you studied and why it matters.
3. **Research Question(s)**: your 1–3 questions (numbered/labeled clearly).
4. **Data Provenance**: where the data came from, who collected it and why, what one row (case / observational unit) represents, and the key variables you use (with units where relevant).
5. **Data Wrangling & Cleaning**: a narrative account of how you imported, tidied, joined, and cleaned the data, and the decisions you made (e.g., dropped rows, recoded values). Describe it in prose; the code itself lives in the appendix.
6. **Exploratory Data Analysis**: your figures and table(s) with the surrounding discussion. This must contain at least three figures, at least one non-trivial table, and at least one figure with 3+ variables.
7. **Findings / Conclusions**: clearly stated answers to your research question(s), tied directly back to the questions in Section 3.

8. **Limitations & Future Work**: what your data or analysis could not address, and what you would do with more time or data.
9. **AI Use Statement**: see the AI policy below.
10. **References**: citations for your data and any outside code or sources.
11. **Code Appendix**: all code, using the `ref.label` snippet above.

Reproducibility

- ☐ Use a GitHub repository to manage the project.
- ☐ Include a README explaining how to reproduce the analysis.
- ☐ The final report must render cleanly from your repo on someone else's machine.

AI policy for the project

AI tools (ChatGPT, Claude, Copilot, etc.) are permitted on all project work. Per the course AI policy:

1. Include an **AI use note** in your final report (1–2 sentences per team member or a paragraph for the team).
2. **You will be asked to walk through your own code at oral checkpoints.**

If a team member can't explain code submitted under their name, that affects their individual project grade regardless of the team's overall result.

Checkpoints

- Final report rubric

FAQs

Can we use a past project from another course? No. That is a violation of academic integrity.

Can we use a project from another *current* course for this one? No.

Can we use data from a research lab one of us works in? Yes, with permission from the lab's PI.

Can we use data from another course's project? Tentatively, yes. Disclose it as part of provenance and check with me early. You may not reuse the same analysis.

Can I use Python? STAT 184 is an R course, and your analysis must be self-contained in a single QMD. If you want to include a small Python chunk for something we haven't covered in R, that's fine.

We're stuck on a topic. Come to office hours. We'll talk through what your team is interested in. Shared hobbies are usually the best starting point.